

UNIT-1

→ Overview of 5G Broadband Wireless Communications

- Mobile communications generations: 1G to 4G
- Rationale of 5G-requirements
- Standardization activities

* 5G

- It is the latest generation
 - of mobile communication technology.
- It is designed to provide :
 - Very high data speed
 - low latency
 - reliable connectivity
- It supports human communication & M/c commⁿ (IoT)
- It is the fifth generation of mbl wireless technology.
- It is developed to meet future commⁿ demands.
- Supports IoT, massive device connectivity
- It is designed for smart & automated systems.
- It supports real-time communication.
- It connects people, devices & machines.
- 5G operates in low, mid & high frequency bands.

Benefits

- Offers ultra-low latency (almost no delay).
- Supports millions of devices simultaneously.
- Supports real-time gaming & VR/AR.
- Provides Very high download & upload speeds.
- Improves n/w capacity.
- Real reliability - error-free ^{comm} 100% accuracy
- High-quality Video Streaming
- Smart city applications
- Efficiency-Performance
- Robotics
- Industrial ~~applic~~ automation.

Limitations

- Large investment (high cost)
- Coverage is limited in rural areas. (Poor coverage)
- Needs new 5G-compatible devices.
- Security & Privacy challenges
(due to many connected devices)
- Old mobile phones can't support 5G.
- 5G n/w setup is expensive.
- Need many towers
- Uses more electricity.

Features	1G	2G	3G	4G	5G
Full Form	First Generation	2nd Generation	3rd Generation	4th Generation	5th Generation
Signal type	Analog	Digital	Digital	D	D
Main service	Voice calls	Voice + SMS	Internet	High-Speed Internet	Smart Services
Data Services Speed	Very low	Low	Medium	High	Very high
Internet Support	No	No	Yes	Yes	Yes
Video calling	No	No	Yes	Yes	Yes
Latency	Very high	High	Medium	Low	Very low
Device Support	Very few	Few	Moderate	Many	Massive devices
Technologies	AMPS	GSM, CDMA	UMTS	LTE	5G NR
Applications	Voice only	Calls, SMS	Browsing	streaming	IoT, AI
Examples	old phones	SMS Phones	3G Mobiles	Smartphones	Smart Cars

Voice quality	Poor	Better	Good	VG	Excellent
Data Support	No	No	Yes	Yes	Yes
Year Introduced	1980	1990	2000	2010	2020
Capacity	Very low	low	medium	high	Very high

* Standardization Activities

(I)

- Standardization is the process of creating common rules & specifications for 5G commⁿ
- It ensures that all devices & n/w^s work together globally.

① ITU (Teaches) (International Telecommunication Union)

- defines global requirements for 5G
- It is known as IMT-2020
- ITU gives instructions

→ It says:

- Speed should be high
- Delay should be low
- More devices should connect

→ It does not build 5G

→ It only gives guidelines

② 3GPP (Engineers) builds technology

→ ~~It~~ develops actual 5G standards

→ Design: 5G architecture, n/w, Radio technology,

→ Actually builds 5G system

→ It creates:

- N/w design
- mbl commⁿ rules.

<u>Release</u>	<u>Meaning</u>	→ imp
Release 15	First 5G	
Release 16	Improved	
Release 17	Advanced.	

③. IEEE ^{helps} → Supports technologies

→ Develops wireless commⁿ standards.

→ works on:

- wifi;
- wireless technologies

→ Supports 5G indirectly

ex: You buy a phone in India

- It works in USA (b/c of standardization rules)

⇒ Standⁿ is the process of setting and common rules for 5G commⁿ

ex: mbl works in all countries

Process

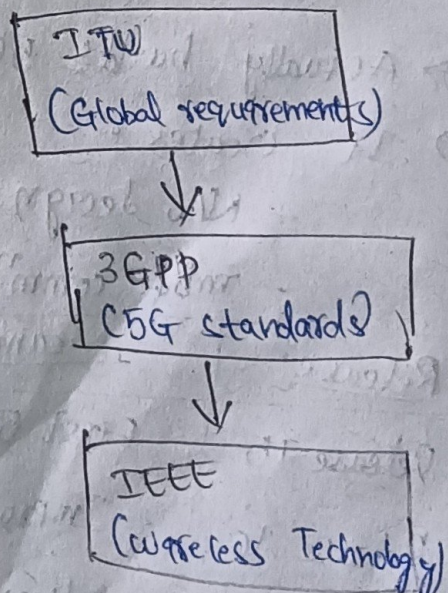
- ITU defines requirements
- 3GPP develops standards
- Testing & Validation
- Approval
- Implementation
- by telecom companies

Need for Standⁿ

- ensures global compatibility
- provides interoperability
(devices work with any n/w)
- maintain quality & performance
- Avoid n/w conflicts
- Reduces cost
- mass production

Features defined by standards

- High data ~~data~~ rate
- low latency
- massive connectivity
- Security
- N/w slicing.



1) High Data Rate Rationale

→ 5G Should provide very high internet speed.

Rationale: - Users watch high-quality videos (4K/8K)

- AR & VR need large data transfer.

- Future ^{(5G) modern} apps need more ^{huge} data speed.

- multiple users use the n/w at the same time.

Req: • Peak data rate up to 20 Gbps

ex: downloading a movie in sec

2) Low Latency (No wait)

• Latency = delay b/w sending & receiving data

→ Very small delay in comm'n.

Rationale:

→ Real-time applications need instant response

→ Delay can cause accidents or errors

applications:

- Remote Surgery

- VR

- AR

- Self-driving cars

- Online gaming

Req: Latency around 1ms

3) Massive Device Connectivity

- Ability to connect a very large no. of devices
- 5G can connect a huge no. of devices

Rationale:

- Smart cities need millions of devices
- 4G cannot handle so many devices.
- IoT devices are rising rapidly.

Req: Support 1 million device per square km

Devices include:

- Sensors
- Smart meters
- Cameras
- Wearables

ex: traffic sensors

4) High Reliability (Trust)

→ Data should reach correctly every time

Rationale

- Some appl^{ns} need error-free commⁿ
- Public Safety & Industrial automation require reliability.

Applications

- Healthcare
- Industrial automation
- emergency services

Req: 99.999% reliability

5) Enhanced Capacity (Crowd handling)

- Ability to handle more data traffic & users.
- N/w should handle more users & more data.

Rationale:

- Dense areas have many users at same time
- Cities are crowded with users.
- events generate heavy traffic

ex: more smartphones, more data usage.

- Stadiums - urban cities

- airports

Req: High n/w capacity with many simultaneous users

- Support high traffic simultaneously

Real-life ex:

- Everyone using internet in a Stadium

6) Energy Efficiency

- Use less Power to operate devices.
- Consume less Power for Comm

Rationale:

- IoT devices run on battery for years
- Charging frequently is not possible
- Reduces operating cost

Req: - Low Power Consumption

- Longer battery life

Benefits

- Longer battery life
- Less power waste
- Environment friendly.

Real-life ex:

→ Sensor working for years on one battery.

→ Support for Mobility ^{moving}

- Reliable commⁿ at high speed movement
- N/w should work even at high speed movement.

Rationale:

- Users travel in trains, cars, metros
- N/w must maintain connection ex:
- Users travel in fast vehicles
 - Train
- Signal should not disconnect.
 - cars
 - metro

Req:

- Support mobility up to 500 km/h

ex:

- Internet working inside high-speed trains
(or) in bullet train

8) Flexible N/w Architecture ^{adjustable}

- N/w should change based on app's need.
- N/w should adapt to different services.

Rationale:

- Different apps need different features.
- one n/w can't treat all services same.
- diff apps have diff needs.
- one n/w must support many use cases.

How achieved:

- N/w slicing

ex:

- IoT slice
- Gaming slice
- Healthcare slice

Benefit

- Better Performance